

Beyond the Slack Barrier: Lean-Verified Bounds and Counterexamples for Erdős 488

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Abstract

For a finite primitive set of integers $G \subseteq \mathbb{N}_{\geq 2}$, let $F_G(n)$ count the positive integers at most n divisible by an element of G , and put $I_G(n) = \sum_{g \in G} \lfloor n/g \rfloor$. We give Lean-verified results concerning the density-doubling inequality in Erdős problem 488. First, the auxiliary estimate $I_G(n) + |G| \leq 2F_G(n)$ holds whenever $n \geq \max G$ and the excess $E = F_G(n) - |G|$ is at most 15. Consequently, the original strict inequality holds at every such lower endpoint and every later upper endpoint. A primitive, collectively coprime example with $E = 16$ shows that 16 is the exact first failure of this auxiliary estimate. This threshold persists at arbitrarily small covered density and arbitrarily large lower endpoints, although the entire resulting sharp family still satisfies the original inequality. We also give explicit counterexamples to Conjectures 4.8 and 6.11 of a specified March 2026 manuscript, prove unbounded tail-sifted density ratios for every fixed primitive ordered pair, and prove unbounded incidence-to-coverage ratios under simultaneous primitivity, collective coprimality, and arbitrary sparsity. The finite classifications are checked by Lean's kernel, with explicit completeness arguments and unchanged state transitions. The unrestricted Erdős problem remains unresolved in this manuscript; global priority for the strengthened formulations is not claimed.

Keywords: sets of multiples; primitive sets; density doubling; finite certificates; Lean; computer-assisted mathematics.

1. Problem, notation, and scope

Write $\mathbb{N} = \{0, 1, 2, \dots\}$. For a finite set $G \subseteq \mathbb{N}_{\geq 2}$, define

$$B_G = \{x \in \mathbb{N} : x \geq 1, \exists g \in G, g \mid x\}, \quad F_G(n) = |B_G \cap [1, n]|,$$
$$I_G(n) = \sum_{g \in G} \left\lfloor \frac{n}{g} \right\rfloor.$$

The formulation of Erdős problem 488 considered here asks whether every nonempty finite $A \subseteq \mathbb{N}_{\geq 2}$ satisfies

$$nF_A(m) < 2mF_A(n) \quad (m > n \geq \max A). \quad (1)$$

This is the sets-of-multiples formulation in the archived formal statement [1]. A set is **primitive** if $a, b \in G$ and $a \mid b$ imply $a = b$. Deleting elements divisible by other elements leaves the same set of multiples and produces a primitive set. Thus primitive sets are a natural setting for the argument.

Assume $n \geq \max G$. Every generator is itself covered, so

$$E_G(n) := F_G(n) - |G| \geq 0.$$

We call $E_G(n)$ the **excess**. It counts covered points other than generators. This is distinct from the signed slack $2F_G(n) - I_G(n) - |G|$. We use **collectively coprime** to mean $\gcd\{g : g \in G\} = 1$, not that every pair of generators is coprime.

The arguments below address two different questions: how far a particular auxiliary estimate proves (1), and why proposed auxiliary assertions can fail without refuting (1). All unbounded statements retain their displayed quantifiers. In particular, an unbounded family of tail-sifted ratios is not a claim about a single fixed tail.

2. Main results

Theorem 1 (the complete excess-at-most-fifteen bound). Let $G \subseteq \mathbb{N}_{\geq 2}$ be finite and primitive, and let $n \geq \max G$. If $E_G(n) \leq 15$, then

$$I_G(n) + |G| \leq 2F_G(n). \quad (2)$$

If $G \neq \emptyset$, then (1) holds for every integer $m > n$.

The inequality (2) also holds for the empty set; only the strict conclusion requires nonemptiness. For arbitrary nonempty finite $A \subseteq \mathbb{N}_{\geq 2}$, let G be its divisibility-minimal subset. The same conclusion holds under $F_A(n) - |G| \leq 15$. Replacing $|G|$ here by the original $|A|$ is not justified.

Theorem 2 (a sharp auxiliary threshold, even under sparsity). Among primitive finite $G \subseteq \mathbb{N}_{\geq 2}$, with $n \geq \max G$, the smallest excess at which (2) fails is 16. More strongly, for every $R, M \in \mathbb{N}$, there are nonempty G and n satisfying these conditions and

$$\begin{aligned} \gcd(G) = 1, \quad RF_G(n) < n, \quad 2F_G(n) < n, \quad M < n, \\ E_G(n) = 16, \quad I_G(n) + |G| = 2F_G(n) + 1. \end{aligned} \quad (3)$$

For each prescribed R, M , 16 remains the least failure excess under all these restrictions. The construction used for (3) nevertheless satisfies the original inequality at its specified lower endpoint for **every positive** upper endpoint, in particular for every $m > n$.

Theorem 3 (unbounded density ratios after a finite prime tail). Fix integers $2 \leq a < b$ with $a \nmid b$. Define

$$S_{a,b|V}(x) = \#\{1 \leq u \leq x : (a \mid u \text{ or } b \mid u), \forall p \in V, p \nmid u\}.$$

For every $K, M \in \mathbb{N}$, $K \geq 1$, there are a nonempty finite set of primes $V \subseteq \{p : p > b\}$ and integers n, m such that

$$\begin{aligned} m > n > M, \quad n &\geq \max(V \cup \{a, b\}), \quad S_{a,b|V}(n) > 0, \\ KmS_{a,b|V}(n) &< nS_{a,b|V}(m). \end{aligned} \tag{4}$$

The entire generator set $V \cup \{a, b\}$ is primitive. The pair a, b is fixed before K, M are chosen; V, n, m may vary.

Theorem 4 (unbounded incidence ratio with collective coprimality). For every $K, R, M \in \mathbb{N}$, there are a nonempty finite primitive set $G \subseteq \mathbb{N}_{\geq 2}$ and $n \geq \max G$ with

$$\gcd(G) = 1, \quad RF_G(n) < n, \quad 2F_G(n) < n, \quad M < n, \quad KF_G(n) < I_G(n). \tag{5}$$

Thus no fixed natural coefficient repairs the auxiliary estimate by replacing its right-hand coefficient 2. The ordinary Archimedean consequence also rules out any fixed real coefficient; the formal theorem is stated with a natural parameter.

An additional exact construction realizes every additive defect $L \geq 2$, with $F = 27L$, $I = 44L$, and $|G| = 11L$, while retaining the same sparsity, coprimality, and endpoint conditions. Unlike (5), this replication construction alone has the fixed ratio $(I + |G|)/F = 55/27$.

3. The auxiliary estimate and its sharp failure

3.1. Why the auxiliary estimate is sufficient

For $G \neq \emptyset$, $n \geq \max G$, and $m > 0$, the union bound and strict floor estimate give

$$F_G(m) \leq \sum_{g \in G} \left\lfloor \frac{m}{g} \right\rfloor \leq m \sum_{g \in G} \frac{1}{g} < \frac{m}{n} \sum_{g \in G} \left(\left\lfloor \frac{n}{g} \right\rfloor + 1 \right).$$

Nonemptiness gives strictness in the last step. Under (2), this is less than or equal to $2mF_G(n)/n$, proving (1). This established reduction appears in the source manuscript [2, Proposition 5.1]; it is independently formalized in `SearchLemmas.lean`.

3.2. A finite refutation of sparse order slack

Conjecture 6.11 of the specific manuscript version [2] proposes a nonnegative total order slack for primitive sets when $2F_G(n) < n$. Its own slack identity makes this equivalent to (2), regardless of the ordering.

Let P be the first sixteen primes, from 2 through 53, and take $G = 2P, n = 220$. Explicitly,

$$G = \{4, 6, 10, 14, 22, 26, 34, 38, 46, 58, 62, 74, 82, 86, 94, 106\}.$$

The certified counts are

$$F_G(220) = 96, \quad I_G(220) = 177, \quad |G| = 16.$$

Hence $2F_G(220) = 192 < 220$, but $I_G(220) + |G| = 193 > 192$. The set is primitive because it is a common nonzero multiple of distinct primes. One can also check the coverage by hand: among $u \leq 110$, precisely 1 and the thirteen primes from 59 through 109 avoid every prime of P . Thus 96 values of $2u$ are covered.

The eleven-generator set

$$G_0 = \{16, 24, 36, 40, 54, 56, 60, 81, 84, 88, 90\}, \quad n_0 = 180 \quad (6)$$

gives the sharper certificate

$$F_{G_0}(180) = 27, \quad I_{G_0}(180) = 44, \quad |G_0| = 11.$$

It is primitive and collectively coprime, $54 < 180$, and $55 > 54$. Thus $E = 16$ and the auxiliary defect is exactly 1. Theorem 1 excludes every smaller excess, establishing Theorem 2' s first claim. The corresponding Lean theorem states the minimum using `IsLeast`.

3.3. The sharp family at arbitrary sparsity

For $k \geq 2$, put

$$H_k = kG_0 \cup \{180k - 1\}, \quad n_k = 180k. \quad (7)$$

The map $x \mapsto kx$ preserves the old covered and incidence counts. The extra generator lies above $n_k/2$, contributes only itself, and is not divisible by k . It is therefore new and uncovered by the old generators. It neither divides nor is divided by an old generator. Consequently,

$$F_{H_k}(n_k) = 28, \quad I_{H_k}(n_k) = 45, \quad |H_k| = 12.$$

A common divisor of $16k$ and $81k$ divides k ; if it also divides $180k - 1$, it is 1. This proves collective coprimality. Taking $k = R + M + 2$ gives all inequalities in (3).

THE FIRST FAILURE IS AT 16

An exact witness for the auxiliary incidence-plus-cardinality bound

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54
55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72
73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108
109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126
127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144
145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160	161	162
163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180

Generator Extra covered point Uncovered

$n = 180$. The original Erdős 488 inequality still holds for this witness at every $m > 180$.

27

covered integers

11

generators

16

extra covered points

$$44 + 11 > 2 \cdot 27$$

Figure 1: The first auxiliary failure at excess 16: orange generators and blue extra covered points at $n=180$. The original Erdős 488 inequality still holds for this witness for every $m > 180$.

These examples do not refute (1). Indeed, exact rational arithmetic gives

$$180 \sum_{g \in G_0} \frac{1}{g} < 48, \quad n_k \sum_{h \in H_k} \frac{1}{h} = 180 \sum_{g \in G_0} \frac{1}{g} + \frac{180k}{180k-1} < 50 < 56.$$

Since $56 = 2F_{H_k}(n_k)$, the rational union bound proves (1) for this whole family and all $m > 0$. The number 16 is the sharp threshold for (2) as a function of excess, not a threshold at which the original conjecture is known to fail.

4. Why the excess-at-most-fifteen proof is finite and complete

Put $q_g = \lfloor n/g \rfloor$, and call g active when $q_g \geq 2$. Its proper-multiple chain is

$$C_g = \{2g, 3g, \dots, q_g g\}.$$

Primitivity ensures that these points are not generators. Their union is exactly $B_G \cap [1, n] \setminus G$, of cardinality E . Writing p for the descending list of active quotients,

$$W(p) := \sum_{q \in p} (q-1) = I_G(n) - |G|,$$

the auxiliary estimate is equivalent to $W(p) \leq 2E$.

Each active generator contributes a distinct doubled point, so there are at most E active generators. Also $q_g - 1 \leq E$. The doubled points, together with fourth and higher dyadic multiples where applicable, give stronger capacity bounds: equality between distinct dyadic multiples would make two generators divide one another. Distinct generators with the same quotient have disjoint proper-multiple chains. To see the latter, a common multiple has two distinct integer coefficients at most that quotient; after reduction their ratio is too far from 1 to allow both generators to have the same integer quotient. The formal proof handles endpoint strictness explicitly.

Further restrictions come from exact pair-intersection capacities, ratio uniqueness, and configurations of three or four quotient values. A minimal auxiliary failure also gives deletion bounds and a connected intersection pattern. These are proved implications about actual integer families, rather than empirical filters.

For a connected family, divide generator coordinates by a chosen smallest generator d . A state contains rational pairs $(r, q) = (g/d, \lfloor n/g \rfloor)$. Its endpoint interval is

$$\max_{(r,q)} qr \leq \frac{n}{d} < \min_{(r,q)} (q+1)r. \quad (8)$$

If a new chain meets an existing proper-multiple point z , its coordinate must be z/j for some integer $2 \leq j \leq q$. This produces finitely many successors. Each transition checks the quotient budget, primitivity, interval feasibility, and the proper-point capacity. A connected actual family has an ordering adding a chain that meets the preceding union; the formal completeness theorem therefore places it on a reachable path.

The certificates are finite tables of states with an initial-state witness and checked successor coverage. A small-core `NoCompletion` certificate proves that no reachable state can realize all requested quotients. A full-profile `Safe` certificate instead proves the needed auxiliary bound at every relevant reachable state. The finite table is useful only after its local checks and the generic reachability theorem have been combined.

For the final excess-15 layer, the complete construction includes 176 profile-enumeration blocks and 43 capacity blocks. Structural reductions and 44 certified cores, containing 3,459 states in total, leave seven profiles. Two final family graphs contain 6,263 states and 20,869 edges. Their complete coverage and safety arguments are supplied to the final theorem. These are counts of certificate states, not counts of distinct integer families or of globally distinct states across all certificates. Earlier excess layers are proved dependencies of the final result.

The search that suggests a profile or graph is outside the trust boundary. The finite list's coverage, every needed edge, and the final implication are checked in Lean. The definitions of `next`, `Bad`, and the original certificate semantics are not weakened to obtain acceptance.

5. Pair-versus-tail: a finite refutation and an unbounded theorem

5.1. A counterexample with all generators prime

For $a = 2, b = 3$, let V be the forty primes from 5 through 181. At $n = 381, m = 1468$, the kernel-checked counts are

$$S_{2,3|V}(381) = 29, \quad S_{2,3|V}(1468) = 224.$$

Thus

$$381 \cdot 224 = 85\,344 > 85\,144 = 2 \cdot 1468 \cdot 29.$$

Every ordering and endpoint condition in Conjecture 4.8 of [2] is met. All generators are distinct primes, so the example even has pairwise coprimality. Below 381 the survivors are precisely the nonunit numbers $2^i 3^j$, of which there are 29. Below 1468 there are 43 such smooth numbers, together with $2p, 3p, 4p, 6p, p > 181$ prime, contributing respectively 88, 51, 31, and 11 numbers. These categories are disjoint.

5.2. Proof of Theorem 3

Choose a sufficiently large t , set $n = 2^{2t}$, and let V consist of all primes p with $b < p \leq n$. Take t large enough that $n > M$ and $V \neq \emptyset$. Every survivor below n has all prime factors at most b . Its prime-exponent vector injects into a finite set with at most

$$(2t + 1)^{b+1}$$

elements. Therefore $S_{a,b|V}(n) \leq (2t + 1)^{b+1}$, while a itself ensures this count is positive.

Let $Q = \prod_{p \in V} p$. An elementary comparison with the product over consecutive odd integers yields

$$Q \leq 2^{t+1} \varphi(Q). \tag{9}$$

For example, the proved Euler-product estimate is $1 \leq 4n(\varphi(Q)/Q)^2$, which implies (9). This estimate requires no prime number theorem.

Set $m = a(n + 1)Q$. For each of $n + 1$ complete residue blocks, the multiples of a indexed by residues coprime to Q survive the tail. Hence

$$S_{a,b|V}(m) \geq (n + 1)\varphi(Q).$$

Combining the two count bounds gives

$$\frac{nS_{a,b|V}(m)}{mS_{a,b|V}(n)} \geq \frac{2^{t-1}}{a(2t + 1)^{b+1}}.$$

A proved elementary exponential-versus-polynomial lemma lets us choose t with $2aK(2t + 1)^{b+1} < 2^t$. This proves (4). Since every tail prime exceeds b , and $a \nmid b$, the combined generator set is primitive.

6. Sparse amplification and incidence growth

6.1. Separated copies

Let $G_0 \subseteq [2, n_0]$ be primitive. For $L \geq 1$ and $N > n_0 L$, define

$$G_{L,N} = \bigcup_{0 \leq i < L} (N + i)G_0, \quad n_* = n_0(N + L).$$

For each $i < L$, $\lfloor n_*/(N + i) \rfloor = n_0$. Moreover, the map

$$(i, u) \mapsto (N + i)u, \quad 0 \leq i < L, \quad 1 \leq u \leq n_0,$$

is injective. Indeed, distinct u, v would force a gap of at least N to be compensated by a term smaller than $n_0 L$. Thus the covered sets of the copies are disjoint, and

$$F_{G_{L,N}}(n_*) = L F_{G_0}(n_0), \quad I_{G_{L,N}}(n_*) = L I_{G_0}(n_0), \quad |G_{L,N}| = L |G_0|. \quad (10)$$

The same separation argument proves primitivity across copies. Applying (10) to (6) yields the exact defect L . If $L \geq 2$, two consecutive scales together with the base elements 16 and 81 force collective gcd one. Increasing N provides arbitrary sparsity and an arbitrary lower-endpoint threshold.

6.2. Proof of Theorem 4

The divergence of the sum of reciprocals of primes is classical and is imported from Mathlib [3]. Choose a finite prime set P , containing 2 and 3, with reciprocal sum sufficiently large, and put

$$N_0 = 2 \prod_{p \in P} p.$$

All selected primes divide N_0 , so there is an exact identity

$$I_P(N_0) = N_0 \sum_{p \in P} \frac{1}{p}.$$

The formal finite-selection argument makes $I_P(N_0) > K(N_0 + 1)$, in particular $I_P(N_0) > K N_0 \geq K F_P(N_0)$. Apply the separated-copies construction with two adjacent scales. Equation (10) preserves the strict incidence inequality. The base primes 2 and 3 and the adjacent scales ensure collective gcd one. Choosing the scale sufficiently large also gives both sparse inequalities and $M < n$.

Bare unboundedness of I/F is already an immediate consequence of prime-reciprocal divergence and $F_P(N_0) \leq N_0$. The formal result recorded here includes the simultaneous

restrictions in (5); no novelty is claimed for the classical divergence theorem or for that elementary unrestricted consequence.

7. Formal verification and reproducibility

The accepted sources use Lean 4.33.1 and Mathlib at commit 0df444a360eaa60ab8c11dca51a86af692955474 [3,4]. Accepted module checks were run with trust level zero (`-t 0`); large computations also use a single Lean worker (`-j 1`). The reported dependencies are only the standard logical axioms `propext`, `Classical.choice`, and `Quot.sound`. The delivered proofs use no `sorry`, `admit`, `native_decide`, or project-specific axioms.

Mathematical result	Principal Lean source and declaration
Theorem 1, auxiliary estimate	<code>ExcessFifteenMain.lean: ExcessFifteen.excess_at_most_fifteen_union_bound</code>
Theorem 1, original inequality	<code>ExcessFifteenMain.lean: ExcessFifteen.erdos488_excess_at_most_fifteen</code>
Reduction of a general input set	<code>PrimitiveCoreBounds.lean: erdos488_core_excess_at_most_fifteen</code>
Explicit sparse slack refutations	<code>ProofsCore.lean, Proofs.lean, SmallSlack.lean</code>
Least failure excess	<code>SharpSlackThreshold.lean: sixteen_is_least_failure_excess</code>
Sharp sparse coprime family	<code>SharpSparseThreshold.lean: arbitrarily_sparse_coprime_sharp_failure</code>
Least excess with all restrictions	<code>SharpSparseLeast.lean: sixteen_is_least_sparse_coprime_failure</code>
The sharp family satisfies the original inequality	<code>SharpSparseOriginal.lean: sharp_sparse_family_original</code>
Finite pair-tail refutation	<code>ProofsSmall.lean</code>
Theorem 3	<code>GeneralPairTail.lean: every_primitive_pair_has_unbounded_tail_density_ratios</code>
Exact additive defect	<code>UnboundedSlackDefect.lean: arbitrary_sparse_coprime_exact_defect</code>
Theorem 4	<code>UnboundedSlackRatio.lean: arbitrary_sparse_coprime_incidence_ratio</code>

Files are rooted at `research/` in the accompanying project. `research/REPRODUCE.md` and `research/verify_bundle.py` give the import-ordered source rebuild. The source archive's manifest fixes the inputs; verification records retain actual exit codes and hashes. Checking a final module against compiled imports is distinguished from rebuilding all imported sources. Selected final entrypoints also have independent agent rebuilds and semantic audits; this does not mean every audit rebuilt the entire dependency graph.

The workspace required a startup adapter because `procs` was unavailable. It supplies the known executable path and calls the original Lean entrypoint; it does not modify proof checking. The original executable and shared-library hashes were compared with the official release. A conventional Linux installation uses the official release without this adapter.

Memory-terminated, interrupted, and superseded attempts are not accepted proofs. They are retained separately from successful checks. Finite certificates are accepted only after their exact statements and relevant imports have actually compiled successfully.

8. Related work, provenance, and the role of computation

The problem statement is pinned to a versioned formal-conjectures source [1], consulted as a specification and never imported as an assumed open theorem. Chojecki’s manuscript [2] proves an excess-at-most-five theorem and proposes the two auxiliary conjectures refuted here. Our references to its numbering mean the PDF dated 20 March 2026, with SHA-256

68639a31a69302089202a71b156425cf5db6f08e7f43fc91ed1869faaeecf5a3

The same manuscript’s Proposition 7.2 already constructs a prime-tail example for a singleton head with density ratio exactly 2. Prime-tail screening therefore has an explicit precursor; it is not claimed as a newly invented idea. The strict counterexamples and the quantified fixed-pair extension are separate statements. Likewise, the standard union bound, elementary scaling, prime-reciprocal divergence, and the integer sequence containing (6) are not claimed as discoveries of this project.

Targeted prior-art checks did not establish global priority. In particular, a relevant August 2026 discussion could not be fully inspected; indexed fragments are not a substitute for its proof or attachments. The defensible claim is that the displayed statements were derived and formally checked in this project, and that Theorem 1 improves the explicit excess range in the inspected March manuscript. Whether all strengthened formulations are new across the literature remains open to external assessment.

The research also included an original-problem counterexample search with 21,600.003879 seconds of recorded effective wall time. Pauses were excluded; parallel CPU time was not substituted for elapsed time. No counterexample was found in the tested inputs. This is a search record, not evidence for an unbounded exclusion theorem. A separate $N = 25$ branch of Erdős problem 686 obtained formal boundary lemmas and finite negative searches, but contributes no theorem to the present paper.

OpenAI Codex agents generated conjectural approaches, search programs, candidate certificates, Lean proof text, mathematical explanations, and this manuscript. Separate agents reviewed several proof interfaces and rebuilt selected entrypoints. Such agent review is not external peer review. The human user set the research objective and authorized publication; no human author identity or institutional endorsement is inferred. Mathematical trust rests on the explicit statements and successful kernel checks, subject to the usual trust in the proof-assistant implementation and its stated logical foundations. Exposition, priority claims, and software packaging still require human scrutiny.

9. Beyond the auxiliary threshold: a pending extension

Status at preparation: the complete excess-at-most-sixteen theorem is not yet claimed. The failure of (2) at excess 16 does not prevent a proof of (1) using a sharper bound. For a normalized state s with upper endpoint $u(s)$, reciprocal sum $R(s)$, and covered-point count $U(s)$,

$$u(s)R(s) \leq 2U(s)$$

is sufficient: the actual normalized endpoint is strictly smaller than $u(s)$. For a nonempty partial state, a separately proved rule bounds the missing reciprocal contribution using the remaining quotients. It explicitly retains the actual equality $F_G(n) = |G| + 16$; it is not an unproved monotonicity claim about numerically promising prefixes.

A complete 626-state certificate already proves the original inequality for the specific active quotient profile $[11, 7, 5, 4, 3, 3, 3, 2, 2, 2, 2]$, under $E \leq 16$, without ambient connectedness or activity assumptions. The broader program has certified five shared graphs covering sixteen remaining profiles and the selected core exclusions. The full raw capacity coverage needed to connect that finite list to every actual input is still pending at this manuscript's preparation. Consequently, none of those components is substituted here for a completed general $E \leq 16$ theorem.

Appendix A. The initial instruction and the actual interaction

The original user instruction is reproduced verbatim below, including its mixed Chinese and English wording. It was a broad research objective, not a supplied mathematical construction or proof.

我有个有趣的主意，首先，你来梳理近代以来数学界所有low hanging fruit，找到20个可用lean解决的问题，但未被AI和人类解决的问题，然后找到问题以后作出可行性分析，在此基础上对一个相对最容易解决的问题发起攻关，允许你使用agent，允许使用一切工具，允许使用lean进行形式化验证，允许使用一切最新资料进行佐证，该问题需具备数学上的开创意义，能够有效推进人类的认知边界，你的结果必须要经过lean的验证，禁止作弊，禁止修改规则，禁止说解决不出来放弃，至少尝试6h。你可以使用multi agent，全自主编排，现在开始，至少运行六个小时，给你完整授权，你可以安装所有你需要的软件，运算资源使用无上限

English translation.

I have an interesting idea. First, survey all the low-hanging fruit in mathematics since modern times and find 20 problems that can be solved using Lean but have not been solved by AI or humans. After identifying the problems, analyze their feasibility and, on that basis, launch an attack on one that is relatively the easiest to solve. You may use agents, all tools, Lean for formal verification, and all the latest materials as supporting evidence. The problem must have pioneering mathematical significance and effectively advance the boundary of human knowledge. Your results must pass Lean verification. No cheating, no changing the rules, and no saying that you cannot solve it and giving up. Try for at least 6 hours. You may use multiple agents with fully autonomous orchestration. Start now and run for at least six hours. You have my full authorization. You may install all the software you need, and there is no upper limit on computational resources.

The agents selected Erdős 488 from the candidate review, designed the constructions and search programs, and developed the Lean proofs. The interaction did not end with this initial instruction: the user later asked the agents to resume after token-limit interruptions and subsequently authorized preparation of a public GitHub project and publication materials.

The workflow therefore should not be described as an uninterrupted, one-click solution. Its recorded search duration is cumulative effective runtime, as explained above.

The instruction's requested scope and ambition are not themselves claims achieved by the paper. In particular, this work supplies a bounded candidate survey, conditional original-problem theorems, and auxiliary counterexamples and strengthenings; it does not establish a complete catalogue of modern open mathematics or solve the unrestricted Erdős 488 problem. The mathematical literature, Lean, and Mathlib remain explicitly credited.

References

1. The Formal Conjectures Authors. *Erdős Problem 488*. Versioned formal statement, commit 8323e878b83fcd7f4a448256069352a265460d75. [Primary source](#).
2. Przemysław Chojecki. *Signed Transport, Pair-Tail Reduction, and Low Layers in an Erdős Density-Doubling Problem*. Manuscript dated 20 March 2026; inspected version identified by the hash above. [Primary manuscript](#).
3. The Mathlib Community. *Mathlib*, commit 0df444a360eaa60ab8c11dca51a86af692955474; in particular `Mathlib/NumberTheory/SumPrimeReciprocals.lean`. [Pinned primary source](#).
4. The Lean Project. *Lean 4*, release v4.33.1. [Official release](#).

The accompanying `refs.bib` supplies these references in BibTeX format. The bibliography records sources used for problem provenance and established ingredients, not an assertion of a complete literature survey.